

shoes that monitor the impact on ground to detect the runner's pace and send information through the Internet to the coach; accessories such as belts, jewellery, gloves and wrist watches with sensing and computing components; head-mounted eyeglasses-based displays; and multimedia systems with wearable cameras, microphones and earphones.

Design challenges

Three significant features differentiate wearable medical devices from other medical devices. First is their wearable nature, which requires that these should be usable in diverse environmental conditions, be comfortable, not noticeable, autonomous, lightweight and compact.

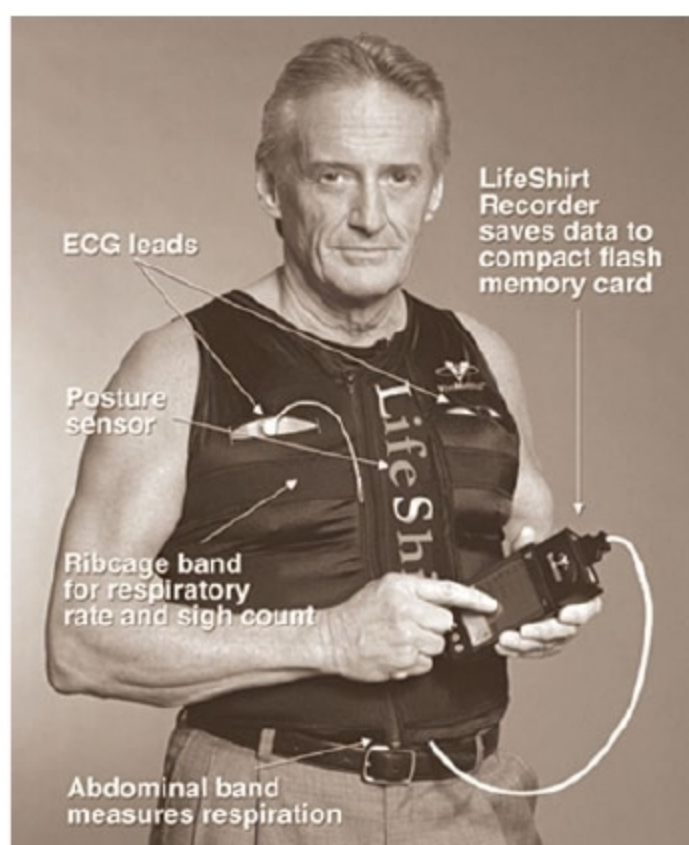
Second is that the patient must be actively involved in their setting up, operation and usage.

Third is that these need to be commercially-viable and economically-affordable for wide acceptance and usage.

These features pose a number of design challenges, as explained below.

Physical challenges. The weight, shape and size play a significant role in making these suitable for their purpose, use and acceptability from the point of view of comfort and fashion. Their relationship and interaction with the human anatomy and the work environment are also important. The following needs to be considered while designing a wearable device:

- The shape, material, texture and colour of the wearable device meets culture and context of use, and the device is comfortable and acceptable to users, easy to set up and operate.
- Weight of the device is distributed evenly on the body, avoiding any effect on natural body movements and balance.
- An appropriate location on the human body to provide a comfortable and steady fit, freedom for movement of joints, and protection from injury and irritants.
- Sensors are non-invasive, avoiding direct skin contact, providing a secure attachment to body without causing any skin irritation, and not



LifeShirt monitors the wearer's vital signs
(Credit: www.virtualworldlets.net)

getting influenced by physiological responses such as vibrations, sweating and prolonged use.

- Devices do not heat up body parts or trap heat.
- Long-term effect of usage is minimum and acceptable.

Technical challenges. These include:

- Data has to be acquired by the device as direct input from the user using keys, touchscreen or speech recognition software, or as output of sensors or through wireless data transfer.
- Analogue inputs have to be converted into digital data before transfer within the device.
- Alterations in contact resistance between the electrodes and skin over time do not affect input data. Environmental factors, noise, ambient temperature and humidity do not influence reliability of acquired data.
- Features such as noise filtering and signal distortion correction, data compression and signal processing for real-time response.
- The design needs to optimise data transfer rates, transmission range, interference between different telecommunication protocols, compatibility problems between devices and power consumption.
- Total power requirement and size of power sources considerably affect total size and weight of the

SMART DIGITAL CLAMPMETER

Power ON-OFF / SELECT (Ω / $\cdot$)

Data Hold

Smart (Auto) Selection of ACA, ACV, DCV

18 SMART

- 3½ Digit 2000 Counts
- Upto 400A AC
- Upto 600V AC / DC
- Upto 20M Ω
- Smart (Auto) Selection of ACA, ACV, DCV
- Hand Free Operation without Rotary Switch
- Simple to Use
- Tool Kit Friendly Size
- Auto Power Off
- Data Hold, Audible Continuity, Low Battery Indicator

Meco Meters Private Limited
 Plot No. EL-60, MIDC Electronic Zone,
 TTC Industrial Area, Mahape,
 Navi Mumbai - 400710 (INDIA)
 Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
 Fax : 0091-22-27673310
 Email : sales@mecoinst.com
 Web : www.mecoinst.com
 Connect With Us :